

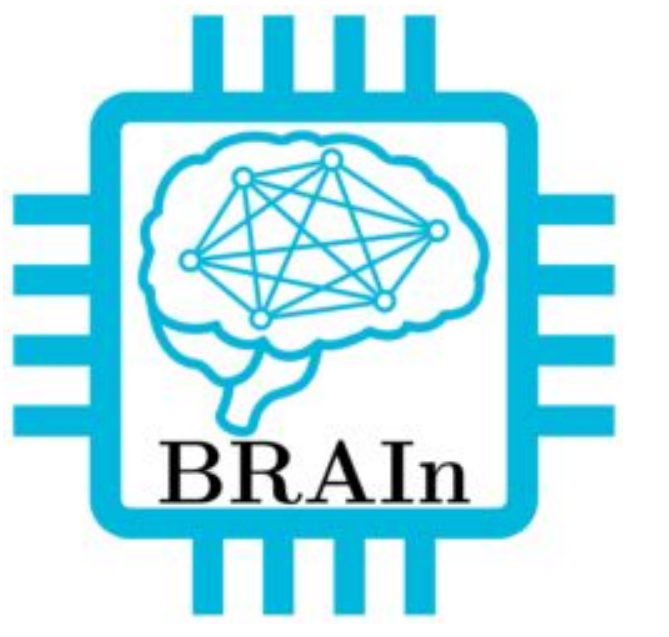


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Spatial Graph Signal Interpolation with an application for Merging BCI Datasets with various Dimensionalities

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Introduction

Motivation

In BCI, there is a lack of big datasets with instead multiple small dataset having different recording setups. Mostly varying by the number and the position of their electrodes.

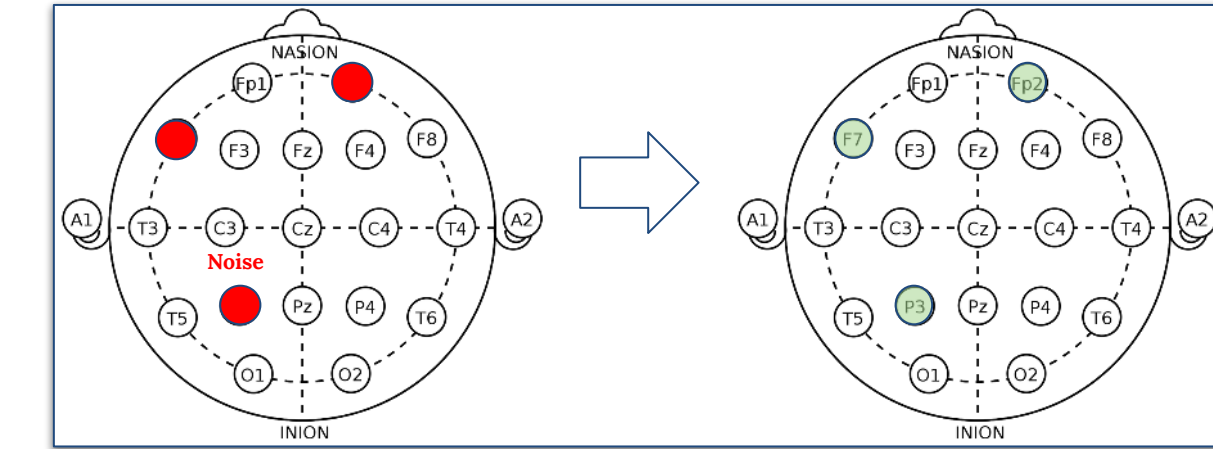


Existing solutions

1 Homogenize BCI datasets

- ❖ Reduce the spatial dimension
➤ ex: PCA, intersection
- ❖ Increase the spatial dimension
➤ ex: Riemannian geometry

2 EEG electrodes interpolation

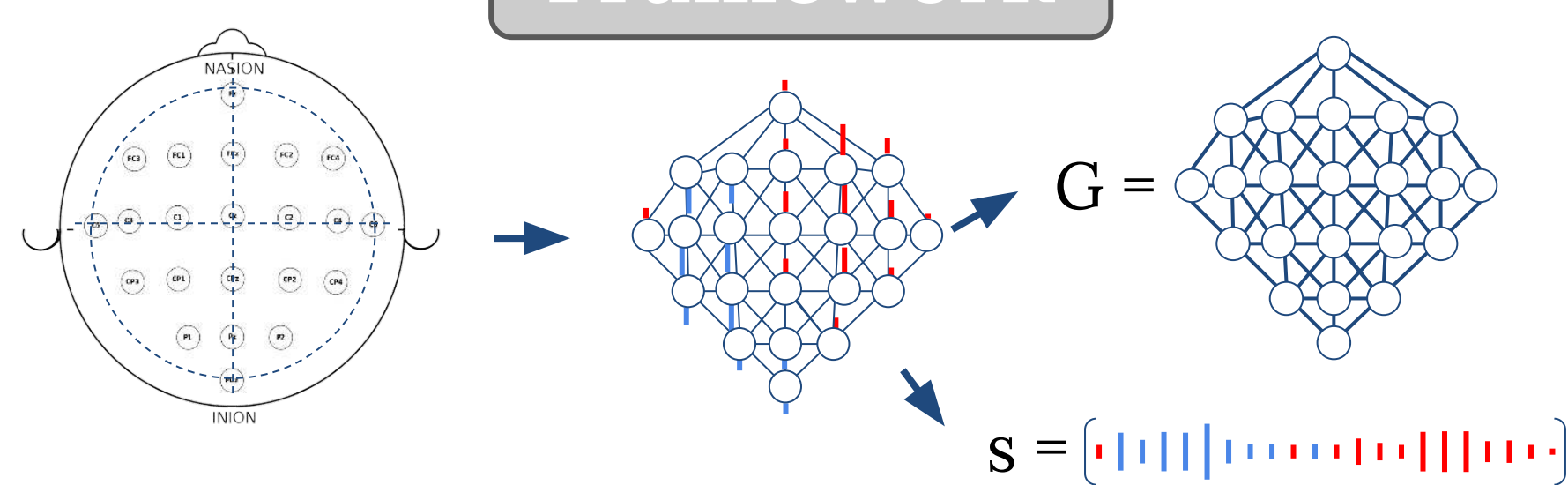


- ❖ Mainly used to recover signal from noisy electrodes using Spherical Spline [1]

Method

We propose to spatially interpolate EEG signal using Graph Signal Processing

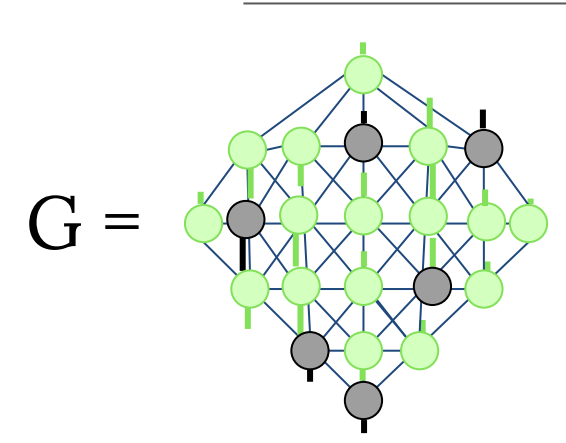
Framework



With D and W , the Degree and Weights matrices of G , we have $L = D - W$, with L the Laplacian.

Smoothness of the signal over the graph is defined by:

$$\sigma(s) = s^T L s = \sum_{i=1}^{|V|} \sum_{j=1}^{|V|} W_{ij} (s_i - s_j)^2$$



$s = [s_1, s_2, \dots, s_n]$ a spatial signal over G
 $s_{\mathcal{M}} = [s_1, \dots, s_m]$ the missing part of s
 $s_{\overline{\mathcal{M}}} = [s_{m+1}, \dots, s_n]$ the observed part of s

Interpolation

1 Finding $s_{\mathcal{M}}$

Interpolation criterion:
smoothness

We minimize:

$$\sigma(s) = s^T L s = \sum_{i=1}^{|V|} \sum_{j=1}^{|V|} W_{ij} (s_i - s_j)^2$$

Using a closed form of $\sigma(s)$ that provides the optimal $s_{\mathcal{M}}$

$$(A) \quad s_{\mathcal{M}} = -L_{\mathcal{M}}^{-1} L_{\mathcal{M}\overline{\mathcal{M}}} s_{\overline{\mathcal{M}}}$$

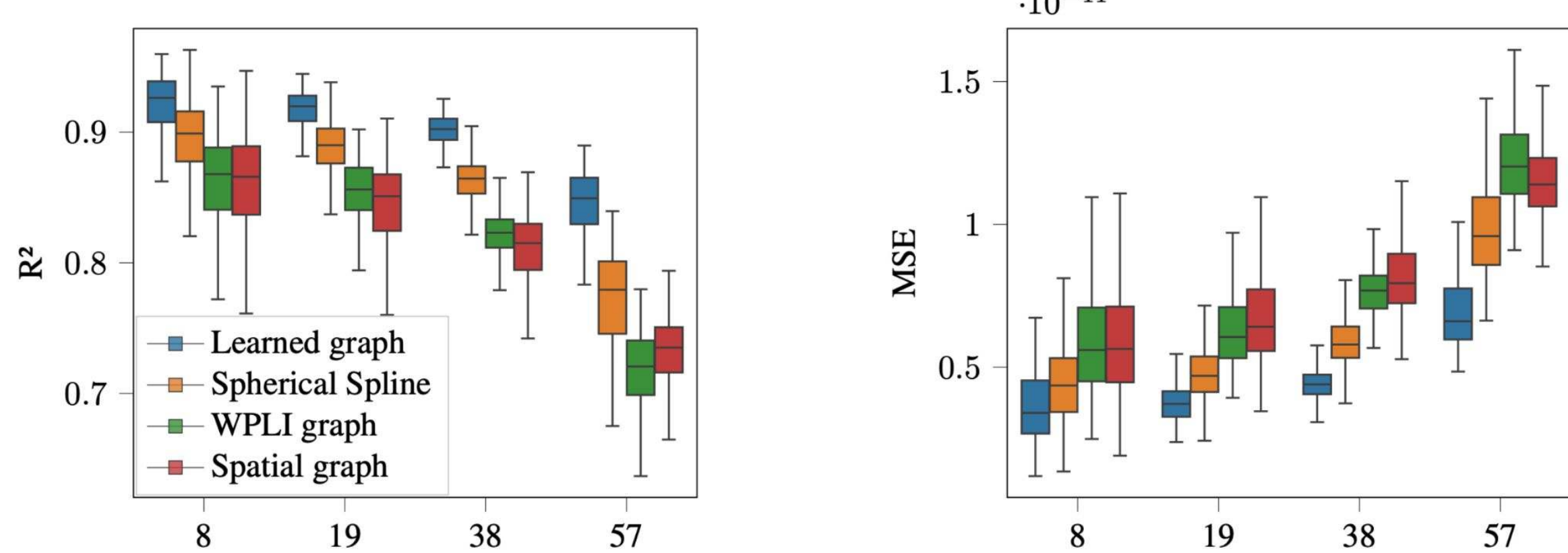
2 Building G

We learn G from real data using gradient descent:

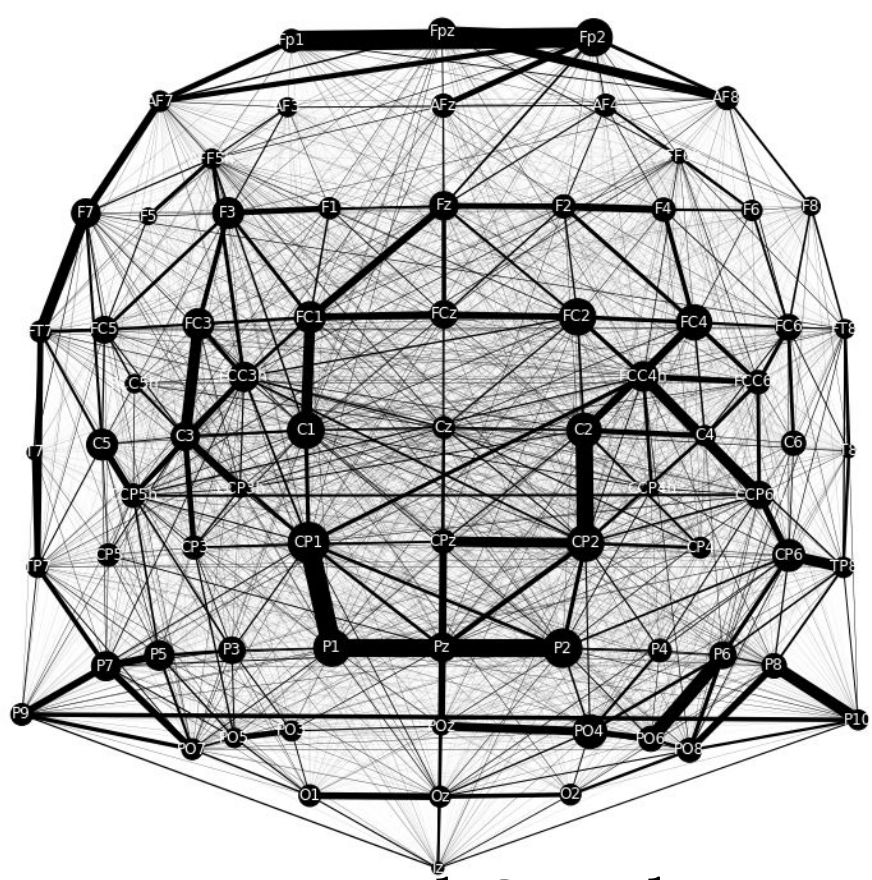
- 1) Initialize a connected graph G
- 2) Create virtual reconstruction problems
- 3) Reconstruct the signal using (A)
- 4) Update G based on the error of reconstruction
- 5) Repeat 2) to 4) until the error is low

Experiments & applications

Signal reconstruction



Reconstruction performances for various number of missing electrodes (among 76 electrodes)



Learned Graph

Nodes size: strength
Edge thickness: weights

Observation:
Cluster in Centro-Fronto-Parietal area on the right side around C4, and symmetrically on the left side around C3.

Merging BCI datasets

Task: Brain decoding setup, inspired from the BEETL Challenge [2]

Sources sets	Cho2017	BCI IV2a	PhysionetMI	Shin or Zhou	
	52 subjects	9 subjects	109 subjects	First 40 trials 29 subjects	First 40 trials 4 subjects
Labels	left-hand right-hand	left-hand right-hand feet tongue	left-hand right-hand feet both hands	left-hand right-hand	left-hand right-hand feet
				Last 40 trials 29 subjects	Last 50 trials 4 subjects
Test sets					

	Shin		Zhou	
	Acc	N	Acc	N
$\cap$	53.2 ± 2.8	2	61.2 ± 2.0	9
Dataset	63.2 ± 2.3	22	56.2 ± 4.8	14
$\cup$	62.3 ± 2.1	76	46.4 ± 2.8	66

Classification accuracy

Conclusion

- ❖ Efficient electrode interpolation technique exploiting Graph Signal Processing tools
- ❖ Illustrated the interest of our method to homogenize datasets
- ❖ Open possibilities to perform cross-dataset experiments in BCI context, such as training neural networks on big merged dataset
- ❖ The interpolation method could be applied for other purposes or in other domains

References

- [1] François Perrin, Jacques Pernier, O Bertrand, and Jean Francois Echallier, "Spherical splines for scalp potential and current density mapping," *Electroencephalography and clinical neurophysiology*, vol. 72, no. 2, pp. 184–187, 1989.
- [2] Xia Wei, A. Aldo Faisal, Moritz Grosse-Wentrup, Alexandre Gramfort, Sylvain Chevallier, Vinay Jayaram, Camille Jeunet, Stylianos Bakas, Siegfried Ludwig, Konstantinos Barmpas, Mehdi Bahri, Yannis Panagakis, Nikolaos A. Laskaris, Dimitrios A. Adamos, Stefanos Zafeiriou, William Duong, Stephen M. Gordon, Vernon J. Lawhern, Maciej Sliwowski, Vincent Rouanne, and Piotr Tempczyk, "2021 beetl competition: Advancing transfer learning for subject independence & heterogenous eeg data sets," *ArXiv*, vol. abs/2202.12950, 2021.

Acknowledgements

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